

HARIRAMAKRRISHNAN RAMACHANDRAN

+353 89 970 6156 • hariramakrrish@gmail.com • Dublin, Ireland • Stamp 1G — Full-Time Work Eligible

PROFESSIONAL EXPERIENCE

Software Engineer — HCL Technologies, Chennai, India

Sep 2021 – Jan 2025

- Developed and implemented **System Verilog** testbenches for complex digital IP blocks, focusing on assertion-based verification to achieve **signoff**.
- Authored **System Verilog Assertions (SVA)** to define and check critical design properties, integrating them into formal verification flows for AI accelerator components.
- Designed and implemented **random test generators** and high-level transactional models within a UVM-based verification environment to stress-test NPU IP functionality.
- Collaborated with **AI architects** and design engineers to translate high-level NPU AI accelerator specifications into detailed verification plans and test cases.
- Scripted **Python** utilities to automate data generation for test case creation and analysis of simulation results, reducing manual effort by approximately **15%**.
- Managed component **release** cycles using Git, participating in code reviews and ensuring adherence to CI/CD practices for verification IP development.

SKILLS

Verification Languages – System Verilog, System Verilog Assertions (SVA)

Verification Methodologies – Formal Verification, Assertion-Based Verification, UVM, Transactional Models

Tools & Environments – EDA Tools (e.g., Synopsys VCS/Formal, Cadence Xcelium/JasperGold), Jenkins, Git, Jira

Programming Languages – Python, C, C++, Shell Scripting, TCL

Hardware Concepts – ASIC, SoC, FPGA, NPU, AI Accelerators, AMBA Protocols, NoC

Cloud & Infrastructure – AWS (EC2, Lambda), Docker

PROJECTS

AI Accelerator IP Verification Environment (HCL Technologies)

- Constructed a **System Verilog**-based verification environment for a digital signal processing IP block, incorporating **BFMs** and scoreboards for functional validation.
- Developed a suite of **SVA properties** to monitor bus transactions and internal state machines, catching over **50** critical design bugs prior to formal signoff.
- Integrated **Python** scripts into the Jenkins CI pipeline to automate test execution and generate coverage reports, improving verification cycle efficiency by an estimated **20%**.

Formal Verification of Control Logic (HCL Technologies)

- Applied **formal verification techniques** using Synopsys VC Formal to prove the correctness of a complex control logic module within a financial operations dashboard system.
- Authored **SVA properties** and functional constraints to formally verify state transitions and deadlock freedom, identifying subtle race conditions missed by simulation.
- Generated detailed reports on formal verification results, mapping found issues to specific design logic and assisting the design team in rapid bug resolution.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin

Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India

2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)